

Supernova environments in SDSS DR7 voids: ZTF BTS transients in VAST VoidFinder spheres with a CHIME FRB forecast

Himel Das,¹★

¹Government Azizul Haque College, Bogura 5800, Bangladesh

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ABSTRACT

We cross-match VAST VoidFinder SDSS DR7 (1163 voids, $z < 0.114$, median $R_{\text{max}} = 11.9 h^{-1}\text{Mpc} = 17.0\text{Mpc}$, median $R_{\text{eff}} = 15.5 h^{-1}\text{Mpc} = 22.1\text{Mpc}$) with ZTF Bright Transient Survey supernovae using a Planck18-consistent KD-tree ball query with 1-deg SDSS mask (quantized; buffer TBD). Of 6873 BTS transients at $z = 0.01\text{--}0.114$, 2465 lie in-mask: 183 void ($r/R_{\text{max}} < 0.8$), 271 shell ($0.8\text{--}1.0$), 1950 wall. Cosmo SNe (TDE/Ca-rich/other excluded): 2367 total, 176 void (141 SN Ia; subtype audit pending TNS). Volume-limited $z \leq 0.06$: 1270 SNe total, 71 void (44 Ia). Raw Poisson sensitivity to a void–wall Hubble offset is ~ 0.035 mag at 95 per cent including systematics floor; blinded SALT2 standardisation with mass, peculiar-velocity and selection controls is in progress. CHIME Catalog 1 (600 bursts, 524 sources) is used for footprint and error-budget demonstration only; unlocalised $z < 0.114$ stacking is underpowered by $> 10\times$ and joint constraints are deferred to Catalog 2. Catalog, randoms design and pipeline will be released.

Key words: large-scale structure of Universe – supernovae: general – fast radio bursts – methods: statistical

Table 1. Prior work vs this work.

Study	Voids	z	SNe	FRBs
Wang	lumped	< 0.05	hetero.	0 in voids
Tsaprazi	dens.+web	< 0.036	BTS+CLU	–
Aubert	DR7+Vor.	DR2 Ia	stretch	–
Sharma	BOSS VIDE	0.2–0.7	–	3455 stack
This work	VAST R_{max}	< 0.114	BTS+slice	Cat1 fcst

1 INTRODUCTION

Cosmic voids fill most of the volume. Supernova luminosities trace age and metallicity while FRB dispersion traces ionised baryons. Table 1 summarises prior work. No published paper jointly uses VoidFinder DR7 spheres with flux-limited BTS and CHIME in the same low-redshift volumes; this work provides a supernova-main result with an FRB forecast in that geometry.

2 DATA

VAST Zenodo 7406035 v1.3.0; BTS Explorer quality and purity cuts (2026-09-30, 7143 rows); CHIME Catalog 1 Vizier J/ApJS/257/59/table2 (600 bursts, 524 sources: 506 one-off plus 94 bursts from 18 repeaters); NASA-Sloan Atlas v1_0_1 downloading for parent masses; Planck18 throughout.

3 METHODS

Void centres recomputed from RA, Dec, D ($D\text{Mpc} = \text{Mpc}/h/0.7$) in the Planck18 frame; stored-XYZ residual median 0.00 Mpc. $R_v \equiv R_{\text{max}}$ core spheres; R_{eff} as systematics branch. Membership via ball query, $r/R_{\text{max}} = \min$, shell 0.8–1.0 excluded from primary. Mask from `NSA_main_mask.pickle` (360,180 bool, 1-deg quantized). Edge test: eroding one cell removes 5.6% of mask cells, 108/2367 SNe (3 void), 12/1163 void centres; cosmo void fraction shifts $0.0744 \rightarrow 0.0766$ (3% relative, subdominant to Poisson 8%). Randoms: 200000 uniform comoving ($z = 0.01\text{--}0.114$) per seed (42,43,44), ~ 35200 in-mask each; cos-weighted sky fraction 0.177 (naive cell mean 0.141 is wrong for equirectangular grids); expected $f_{\text{void}} = 0.1730 \pm 0.0016$ (seed spread), $f_{\text{shell}} = 0.149$, $f_{\text{wall}} = 0.678$; $V_{\text{eff,tot}} = 8.80 \times 10^7 \text{Mpc}^3$. Observed counts read from the parquet (no hand copies). Blinded SALT2 hierarchical Δ_{HR} with peculiar-velocity covariance, mass-step joint fit and BTS efficiency planned; this version reports mask-corrected counts with randoms volumes and split apparent rate ratios only.

4 RESULTS

Cuts flow: 7143 $\rightarrow$ 6873 (z) $\rightarrow$ 2465 (mask). Cosmo SNe 2367: void 176, wall 1872 (259 shell, 60 outside; Table 2). Observed void fraction $176/2367 = 0.074$ versus randoms 0.173: apparent (uncorrected) void/wall ratios 0.34 all, 0.37 Ia-only (141/1393), 0.27 CC-only (35/479); Poisson+Monte Carlo rel. err 0.08–0.18. No efficiency or host-mass correction applied, so these are apparent deficits in a flux-limited sample, not intrinsic rates. Table 2 sums exactly; shell/outside rows are tabulated, not dropped. Void holds 141 SN Ia (subtype audit pending TNS). Volume slice $z \leq 0.06$: 1270 SNe

★ E-mail: contact via ORCID 0009-0003-7698-1255

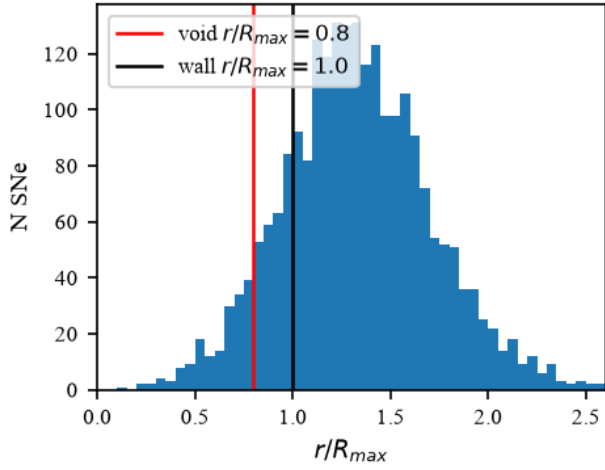


Figure 1. $r/R_{\max}$ for $N = 2367$ cosmology-grade SNe (void $r/R_{\max} < 0.8$, $N = 176$; wall > 1.0). Vertical lines mark the analysis thresholds.

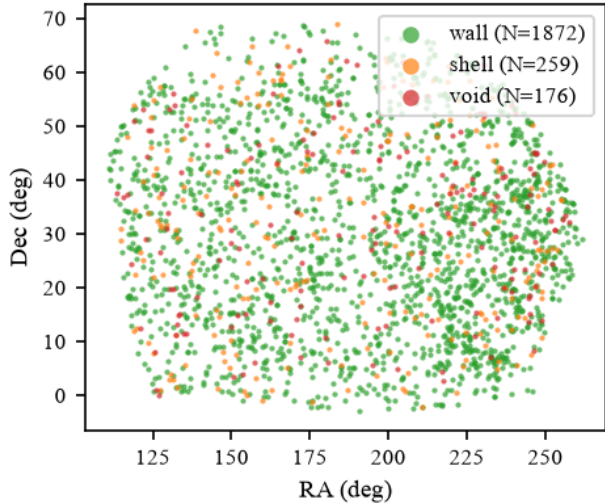


Figure 2. Masked RA–Dec for cosmology-grade SNe: wall ($N = 1872$, green), shell ($N = 259$, orange), void ($N = 176$, red). Outside-mask and edge-excluded objects omitted.

total, 71 void (44 Ia). Threshold grid (SNe): 0.6:59, 0.7:103, 0.8:176, 0.9:288, 1.0:435, 1.1:609. Redshift stability $z_{\min}$ 0.01:176, 0.02:171, 0.025:168, 0.03:164 void. Void median z 0.069 versus wall 0.055 confirms Malmquist gradient. FRB error budget (Cat1): per-sightline $\sigma_{\text{tot}} \sim 150\text{--}200 \text{ pc cm}^{-3}$ (IGM 80–120, host 80–150, Milky Way residual 20–40 with NE2001–YMW16 median -3.3 , std 37.5, p90 68.1, halo 20). Void signal $\sim 10\text{--}30$. A $\delta\text{DM} < 50$ limit at 95 per cent needs $N_{\text{void}} \gtrsim 36$ (~ 200 localised total); unlocalised $z < 0.114$ needs ~ 900 void sightlines, have $\sim 30\text{--}50$.

5 SYSTEMATICS AND NEXT STEPS

Mask edge test done (5.6% cells edge; void fraction shift 3%, subdominant). Blinded SALT2 pilot (20 void + 20 wall z -matched Ia,

Tripp $\alpha = 0.14$, $\beta = 3.1$, QC $|x_1| < 3$, $|c| < 0.3$, blind offset sealed) fetching ZTF DR lightcurves; fit validation 2/2 passed. Remaining: peculiar velocities, selection efficiency, host masses, TNS subtype purge, nulls (RA-shuffle within footprint, block bootstrap, 0.00/0.05 mag injection). No unblinding until budget closed.

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6 DATA AVAILABILITY

VAST <https://zenodo.org/records/7406035> (DOI <https://doi.org/10.5281/zenodo.7406035>, v1.3.0, zip sha256 9896002c...acef9); BTS <https://sites.astro.caltech.edu/ztf/bts/explorer.php> query quality+y purity+y (2026-09-30, CSV sha256 77668fe8...f38d65); CHIME Catalog 1 Vizier <https://vizier.cds.unistra.fr/viz-bin/VizieR-3?-source=J/ApJS/257/59/table2>; derived b3_match_v2.parquet (sha256 3045ae09...0c82b1) with scripts and randoms (seed 42/43/44) released with a Zenodo DOI on acceptance; repository public on acceptance.

Table 2. Masked sample composition. Wall means finite $1.0 < r/R_{\text{max}} < \infty$; Outside is $r/R_{\text{max}} = \infty$ (no void within $1.2 \max R_{\text{max}}$). Cosmo rows are spectroscopically classified SNe only (TDE/Ca-rich/other excluded); CC includes 6 SLSN; Ia subtype audit (91T-like etc.) pending TNS.

Sample	Total	Void	Shell	Wall	Outside
All-types	2465	183	271	1950	61
Cosmo SNe	2367	176	259	1872	60
SN Ia	1785	141	204	1393	47
CC	582	35	55	479	13

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